

Question 16**1 mark**

If the point $(4, -3)$ lies on the graph of $f(x)$, which point must lie on the graph of $2f(2x)$?

- a) $(8, -6)$ b) $(2, -6)$ c) $\left(8, -\frac{3}{2}\right)$ d) $\left(2, -\frac{3}{2}\right)$

Question 17**1 mark**

The graph of $y = \log_2(2x + 6)$ intersects the graph of $y = 4$ at:

- a) $x = -1$ b) $x = 1$ c) $x = 5$ d) $x = 14$

Question 18**1 mark**

Given the point $A(-3, 5)$ on the terminal arm of an angle θ , identify the value of $\cot \theta$.

- a) $-\frac{3}{5}$ b) $-\frac{5}{3}$ c) $-\frac{4}{5}$ d) $-\frac{5}{4}$

Question 19

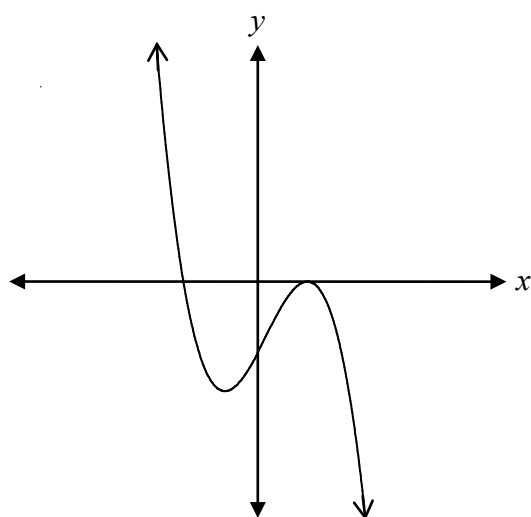
1 mark

The graph of $y = \left(\frac{1}{2}\right)^x$ compared to the graph of $x = \left(\frac{1}{2}\right)^y$ is a:

- a) reflection in the x -axis
- b) reflection in the y -axis
- c) reflection in the line $y = x$
- d) reciprocal function

Question 20

1 mark



Given the above graph of a polynomial function, which one of the following statements can be true?

- a) The function has a degree of 4 with a positive leading coefficient.
- b) The function has a degree of 4 with a negative leading coefficient.
- c) The function has a degree of 3 with a positive leading coefficient.
- d) The function has a degree of 3 with a negative leading coefficient.

Question 21

1 mark

Given that $(x+3)$ is a factor of polynomial $P(x)$, which of the following is true?

- a) $P(-3) = 0$ b) $P(0) = -3$ c) $P(0) = 3$ d) $P(3) = 0$

Question 22

1 mark

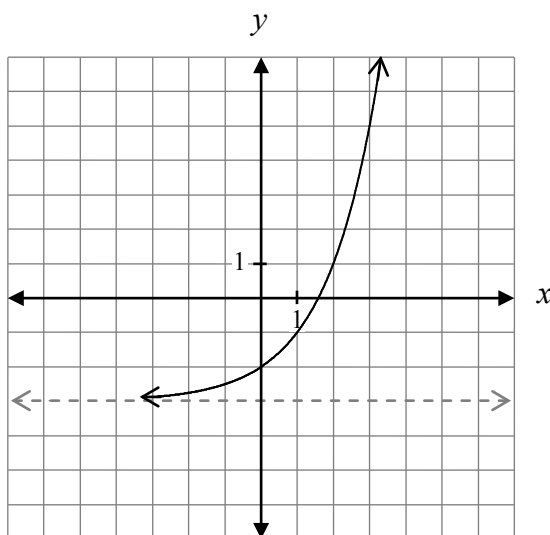
Which of the following is a reasonable estimate for the value of $\log 350$?

- a) 2 b) 2.5 c) 2.8 d) 3

Question 23

1 mark

The graph of the function $f(x)$ shown below is best described by the equation:

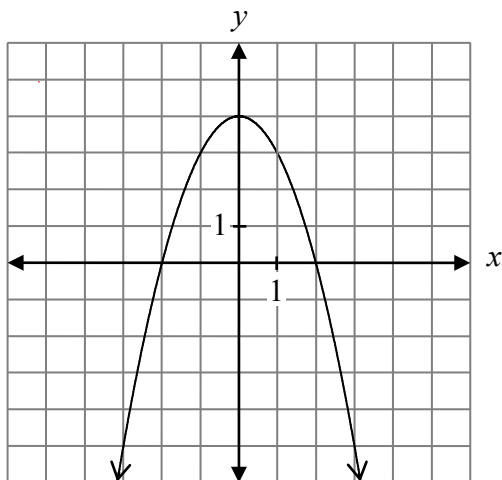


- a) $f(x) = 2^{x+3}$ b) $f(x) = 2^x + 3$ c) $f(x) = 2^{x-3}$ d) $f(x) = 2^x - 3$

Question 24

1 mark

Given the graph of $y = f(x)$, what is the domain of $\sqrt{f(x)}$?



- a) $x \in \mathbb{R}$ b) $-2 \leq x \leq 2$ c) $x \leq -2$ or $x \geq 2$ d) $0 \leq x \leq 4$

Question 25

1 mark

Solve:

$$e^{\ln(5-x)} = 7$$

- a) -2 b) $-\ln 2$ c) $\ln 7 - \ln 5$ d) $\frac{7}{5}$

Question 26

2 marks

118

One of the factors of $P(x) = x^3 - kx^2 - 7x + 10$ is $(x - 2)$.

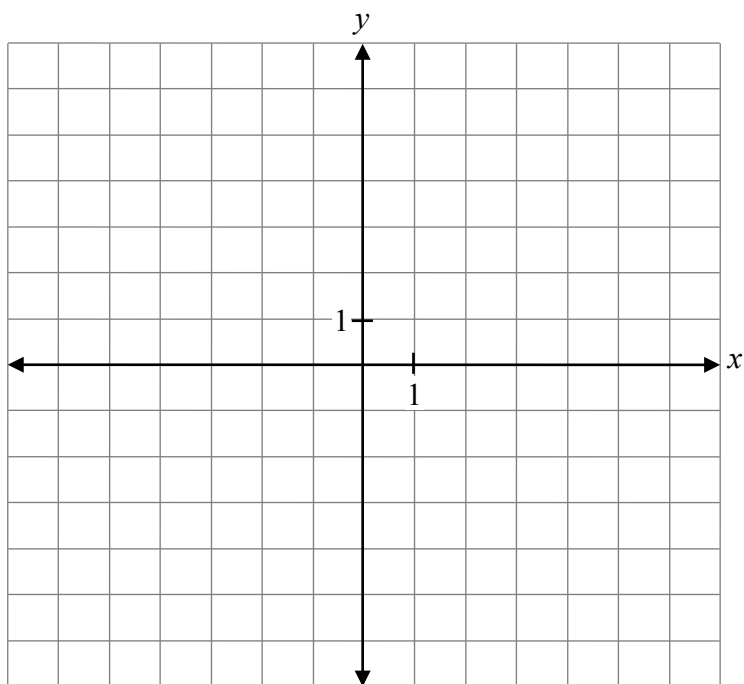
Find the value of k .

Question 27

a) 3 marks b) 1 mark

119
120

- a) Sketch the graph of the function $y = \sqrt{-x} + 1$.



- b) Determine the value of x when $y = 3$.

Question 28

3 marks

121

Solve the following equation:

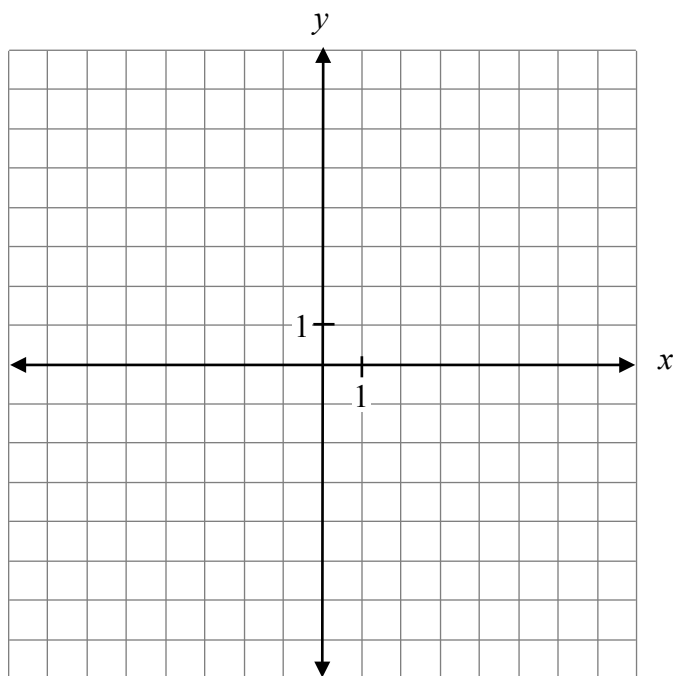
$${}_nP_2 = {}_nC_3$$

Question 29

3 marks

122

Given $f(x) = x^2 - 1$ and $g(x) = \sqrt{x+1}$, sketch the graph of $y = f(g(x))$ and state its domain.



Domain: _____

Question 30

1 mark

123

Write the equation of the horizontal asymptote for the function $f(x) = \frac{x-3}{x-2}$.

Question 31

1 mark

124

The x -intercept of $f(x)$ is 4 and the x -intercept of $g(x)$ is 4.

Benjamin concludes that the x -intercept of $f(x) + g(x)$ is 8.

Do you agree with Benjamin? Justify your answer.

Question 32

4 marks

125

Solve the following equation:

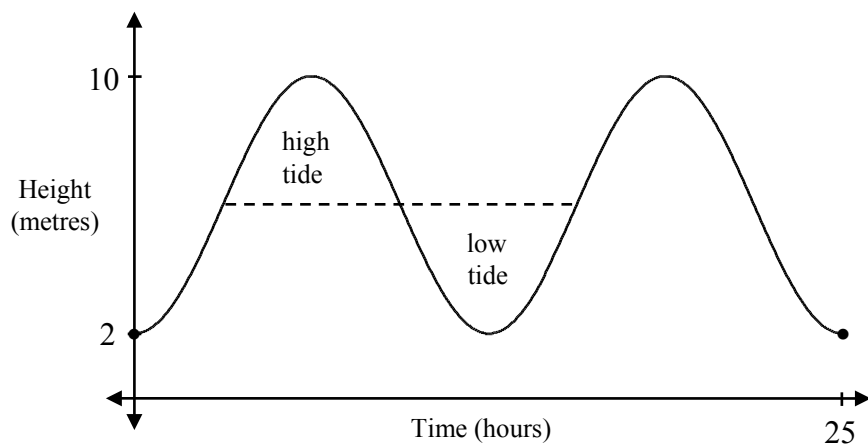
$$2\log_4 x - \log_4 (x + 3) = 1$$

Question 33

a) 1 mark b) 2 marks

126
127

The following graph represents tidal levels in the Bay of Fundy over a 25-hour period.

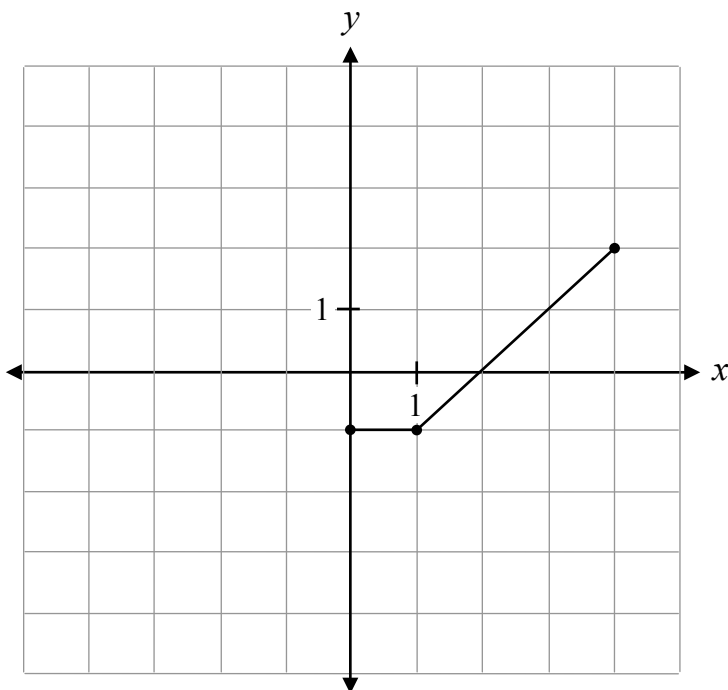


a) What is the average height of the water?

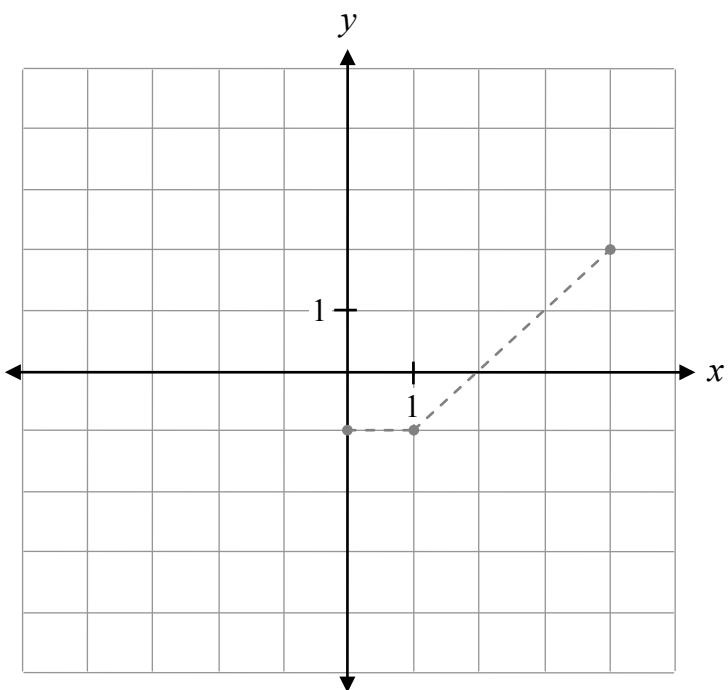
b) What is the period of the graph above?

Explain what the period represents in this situation.

Given the graph of $y = f(x)$ below,



sketch the graph of $y = \sqrt{f(x)}$.



The graph of $f(x)$ has already been drawn for your reference.

No marks will be awarded for the graph of $f(x)$.

Question 35

1 mark

129

When $P(x)$ is divided by $x - 3$, it has a quotient of $2x^2 + x - 6$ and a remainder of 4.

Determine $P(x)$.

Question 36

2 marks

130

Identify the domain and range of the following function:

$$f(x) = \frac{3}{x^2 + 1}$$

Evaluate:

$$\csc\left(\frac{11\pi}{6}\right) + \sin^2\left(-\frac{3\pi}{4}\right) + \cos\left(\frac{23\pi}{3}\right)$$

Question 38

2 marks

132

Evaluate the coefficient of the term containing x^3 in the expansion of $(1+x)^7$.

Justify your answer.

Question 39

1 mark

133

Which of the following equations could be solved without the use of logarithms?

Without actually solving the problem, explain your choice.

$$4^x = 10^{3x+1} \quad \text{or} \quad \left(\frac{1}{3}\right)^{2x+1} = 27^{4x-1}$$

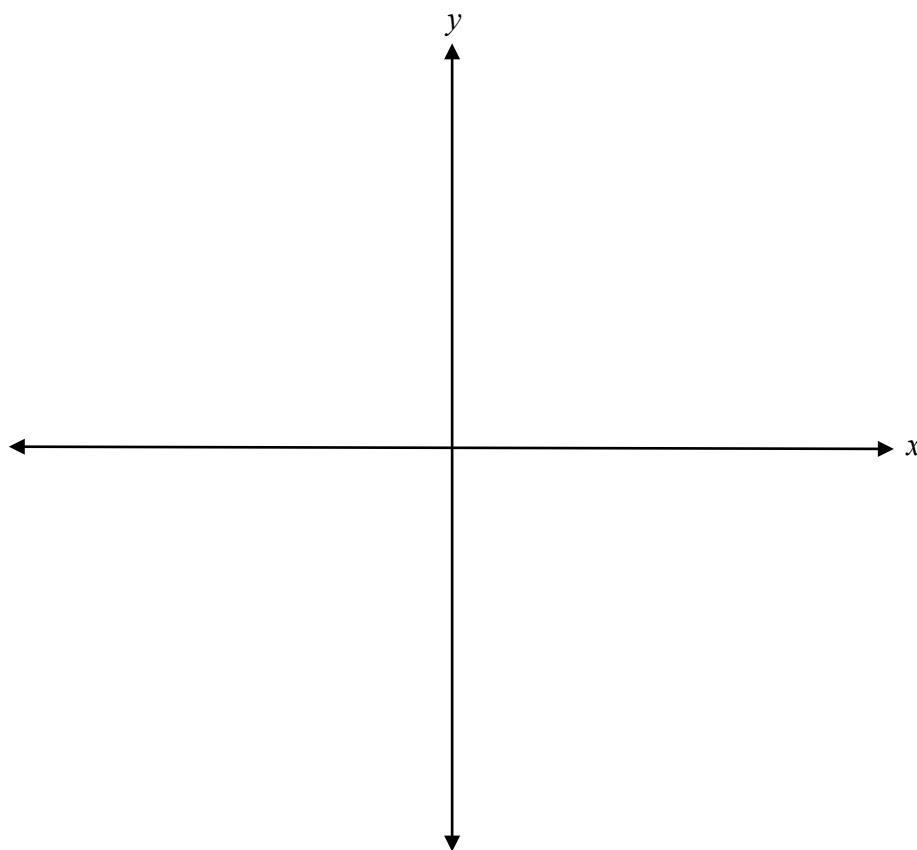
Question 40

4 marks

134

Sketch the graph of $y = x^3 + x^2 - 5x + 3$ given that one of the x -intercepts is 1.

Identify the x -intercepts and y -intercept.



Question 41**1 mark**

135

If $f(x) = \frac{1}{x-2}$ and $g(x) = x - 2$, what is the domain of $f(x) \cdot g(x)$?

Question 42**2 marks**

136

Given $f(x) = (x+1)^2$ for $x \leq -1$, write the equation of $y = f^{-1}(x)$.

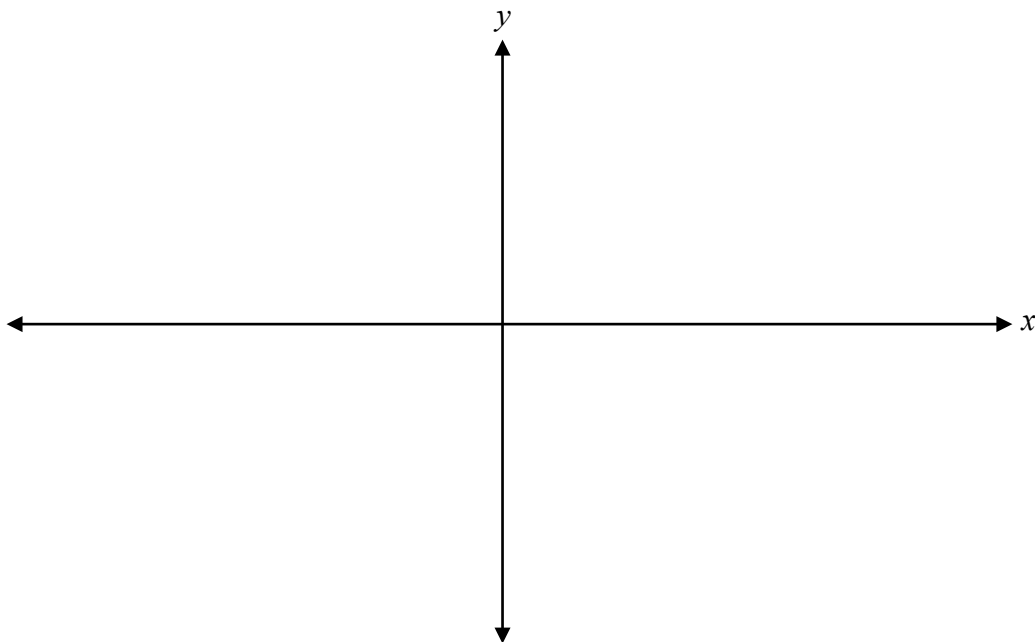
Question 43

4 marks

137

Sketch a graph of at least one period of the function $y = 5 \sin[\pi(x + 1)]$.

Clearly indicate the x -intercepts.



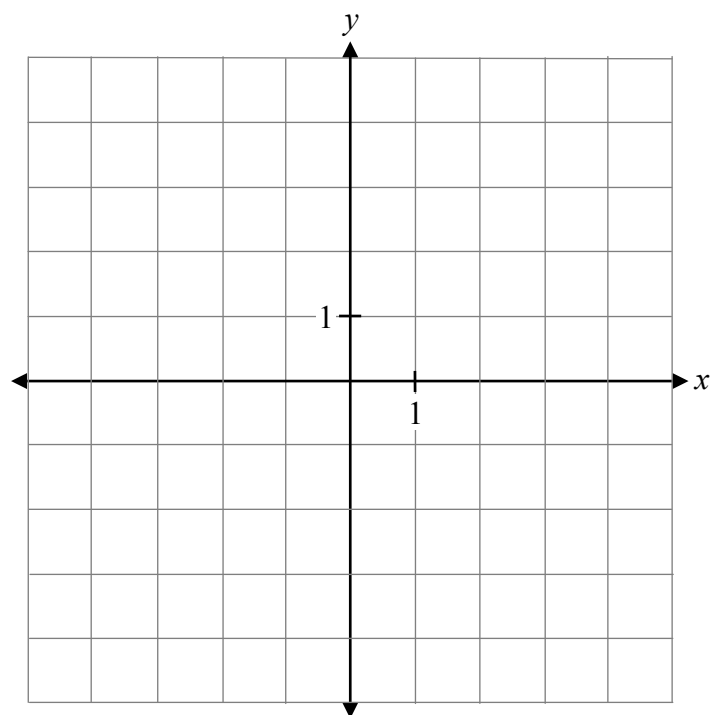
Question 44

4 marks

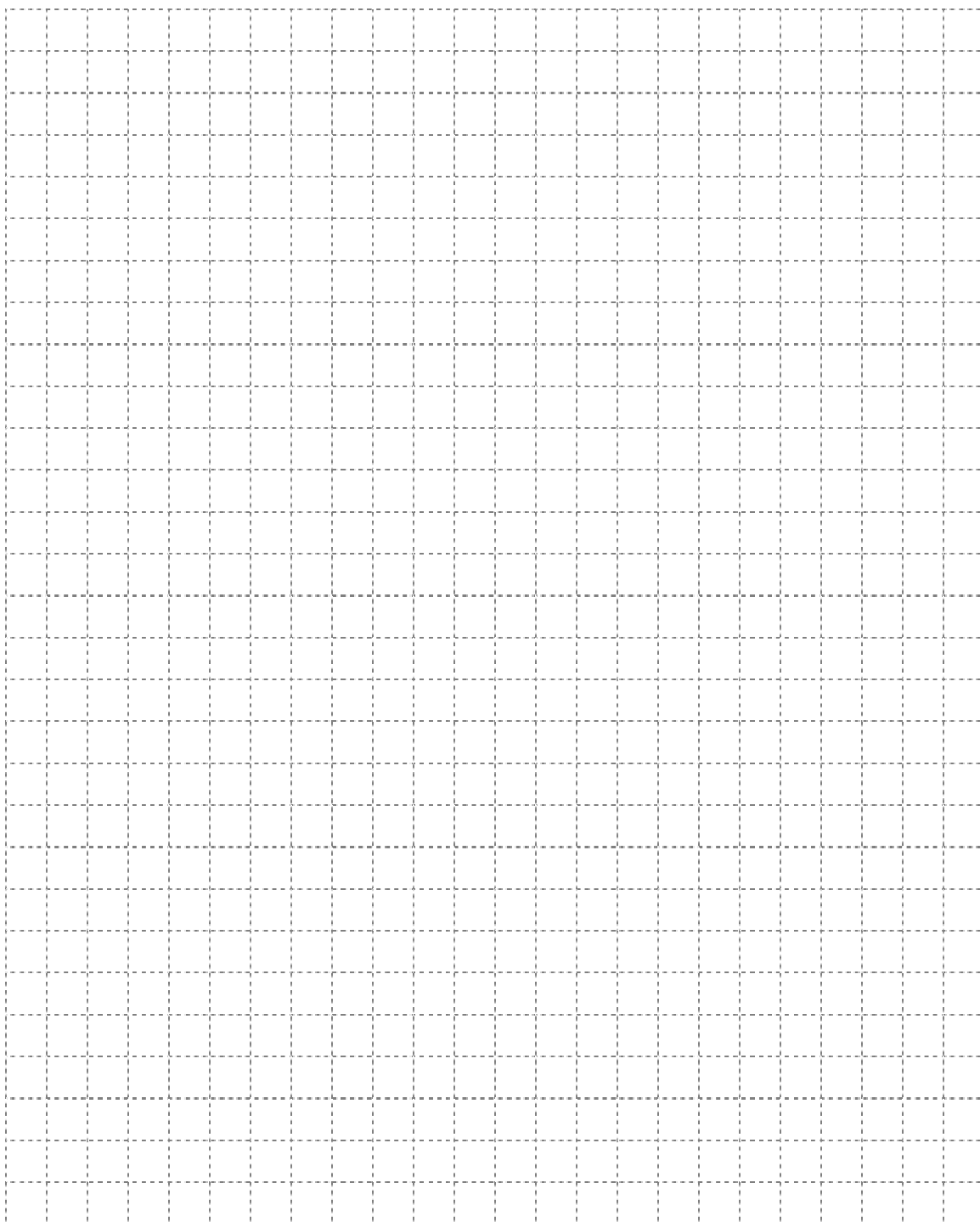
138

Sketch the graph of the following function:

$$f(x) = \frac{x-2}{(2x-3)(x-2)}$$



No marks will be awarded for work done on this page.



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